NETFLIX
CONTENT ANALYSIS

An end-to-end data analytics project using Python, SQL and Power BI

Prepared by	Seher Younis Wani
Project type	Exploratory data analysis and interactive dashboard
Tools	Python • Pandas • NumPy • Matplotlib/Seaborn • MySQL • Power BI

Project purpose

This project explores a Netflix titles catalogue to understand its content mix, release patterns, ratings, countries, genres and selected content details. Python is used for data inspection and preparation, MySQL for structured querying, and Power BI for communicating findings through an interactive report.

1. Project Overview

The goal is to turn a raw catalogue dataset into an organised analysis and an easy-to-explore dashboard. The workflow connects data preparation, exploratory analysis, SQL querying and visual storytelling.

Project workflow

1. Load and inspect the dataset in Jupyter Notebook.
2. Check data types, missing values and possible inconsistencies.
3. Clean selected fields and explore the data using Pandas and NumPy.
4. Load the data into a MySQL database and answer business-style questions with SQL.
5. Build a multi-page Power BI dashboard to summarise the catalogue and allow users to explore titles.

Questions this project can answer

• How many catalogue records are movies versus TV shows?

• Which release years have the most titles in the dataset?

• How does the catalogue vary by rating, country and listed genre?

• What do movie-duration and TV-season distributions look like?

• How can a user select a title and inspect its available metadata?

Notebook: dataset loaded and inspected

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('netflix_titles.csv')
```

[2]:
```
df.head()
```

[2]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020	PG-13	90 min	Documentaries	As her father nears the end of his life, filmm...
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, TV Dramas, TV Mysteries	After crossing paths at a party, a Cape Town t...
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...	To protect his family from a powerful drug lor...
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021	TV-MA	1 Season	Docuseries, Reality TV	Feuds, flirtations and toilet talk go down amo...
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021	TV-MA	2 Seasons	International TV Shows, Romantic TV Shows, TV ...	In a city of coaching centers known to train I...

[3]:
```
df.describe(include = 'all')
```

[3]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	listed_in	description
count	8807	8807	8807	6173	7982	7976	8797	8807.000000	8803	8804	8807	8807
unique	8807	2	8807	4528	7692	748	1767	NaN	17	220	514	8775
top	s1	Movie	Dick Johnson Is Dead	Rajiv Chilaka	David Attenborough	United States	January 1, 2020	NaN	TV-MA	1 Season	Dramas, International Movies	Paranormal activity at a lush, abandoned prope...
freq	1	6131	1	19	19	2818	109	NaN	3207	1793	362	4
mean	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2014.180198	NaN	NaN	NaN	NaN
std	NaN	NaN	NaN	NaN	NaN	NaN	NaN	8.819312	NaN	NaN	NaN	NaN
min	NaN	NaN	NaN	NaN	NaN	NaN	NaN	1925.000000	NaN	NaN	NaN	NaN
25%	NaN	NaN	NaN	NaN	NaN	NaN	NaN	2013.000000	NaN	NaN	NaN	NaN

2. Dataset Description

The Netflix titles dataset is a catalogue-level dataset. Common fields include show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in and description.

Important fields

Field	Meaning / use
show_id	Record identifier; useful for joining or checking duplicate records.
type	Whether a record is a Movie or TV Show.
title / release_year	Title name and original release year.
country / listed_in	Country and genre/category metadata; values may contain multiple comma-separated labels.
rating / duration	Content rating and runtime or number of seasons, depending on type.
date_added / description	Date added to the catalogue and text description, when available.
Director / cast	Director name and Actor name.

Dataset columns and data types

```
[5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

3. Python Data Cleaning and Preparation

Python was used in Jupyter Notebook to inspect and prepare the data. Pandas supports tabular operations such as reading files, selecting columns, checking missing values and grouping records. NumPy supports numerical operations, while Matplotlib and Seaborn can be used to visualise distributions and relationships.

Known cleaning example

The following step fills missing values in the country, director, cast, duration and rating columns with the label “Unknown”.

```
df[["country", "director", "cast", "duration", "rating"]] = df[["country", "director", "cast",
"duration", "rating"]].fillna("Unknown")
```

Good checks to document

• Inspect missing values before and after cleaning.

• Check data types and confirm date/duration fields are handled appropriately.

• Review duplicates and decide how to handle them based on the dataset and project goal.

• Keep the original dataset unchanged and work on a prepared copy where practical.

Missing-value check before cleaning

```
[13]: df.isnull().sum().sort_values(ascending=False)

[13]: director        2634
      country          831
      cast             825
      date_added        10
      rating             4
      duration           3
      show_id            0
      type               0
      title              0
      release_year       0
      listed_in          0
      description        0
      dtype: int64
```

Cleaning code and result

```
[16]: df[["country", "director", "cast", "duration", "rating"]] = df[["country", "director", "cast", "duration", "rating"]].fillna("Unknown")

[17]: df.isnull().sum().sort_values(ascending=False)

[17]: date_added      10
      show_id          0
      type             0
      title            0
      director         0
      cast             0
      country          0
      release_year     0
      rating           0
      duration         0
      listed_in        0
      description      0
      dtype: int64
```

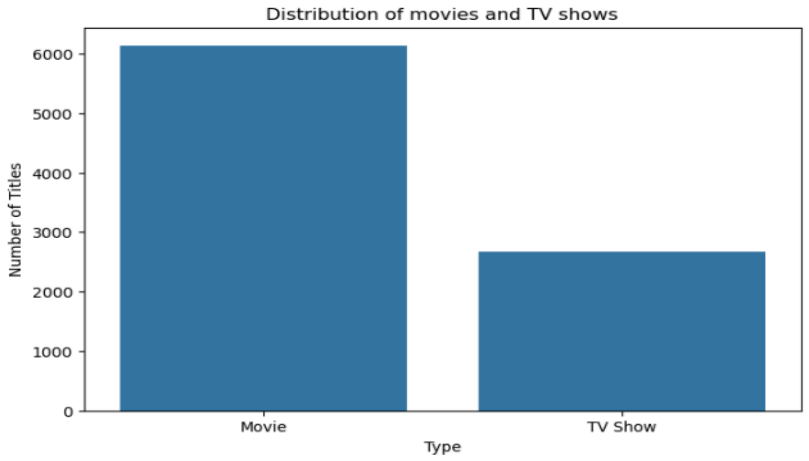
4. Exploratory Data Analysis (EDA)

Exploratory data analysis helped reveal patterns and data-quality issues before building a dashboard.

EDA chart: Movie vs TV Show

```
[23]: plt.figure(figsize=(8, 5))

sns.countplot(data=df, x="type")
plt.title("Distribution of movies and TV shows")
plt.xlabel("Type")
plt.ylabel("Number of Titles")
plt.show()
```



The distribution of Movies and TV Shows on Netflix shows that Movies make up a larger proportion of the total titles than TV Shows.

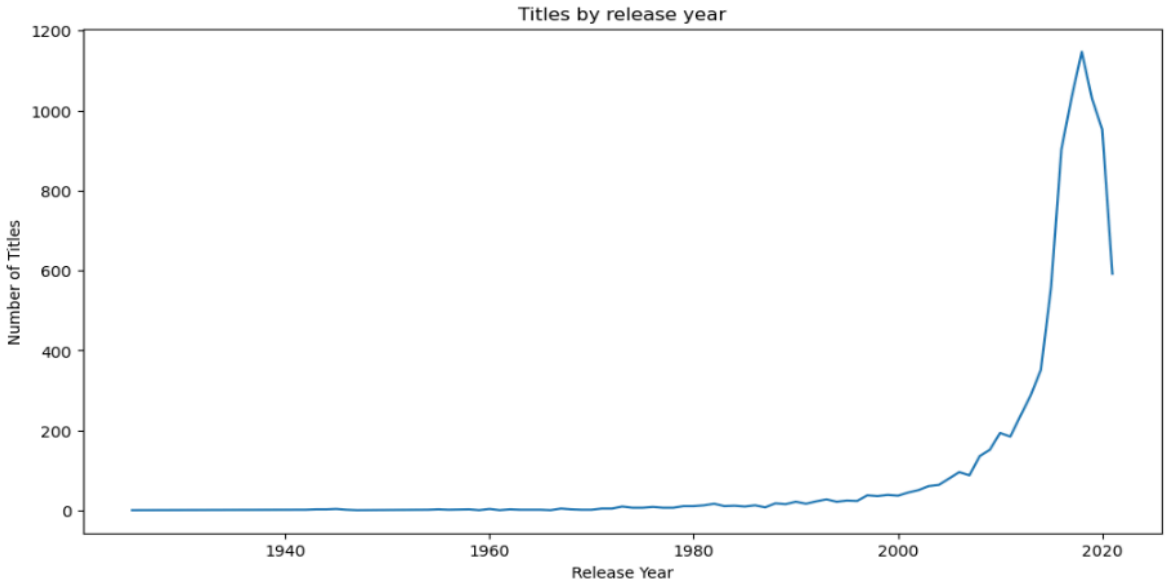
EDA chart: Release-year trend

```
[24]: year_count = df["release_year"].value_counts().sort_index()

[25]: plt.figure(figsize=(12, 6))

plt.plot(year_count.index, year_count.values)

plt.title("Titles by release year")
plt.xlabel("Release Year")
plt.ylabel("Number of Titles")
plt.show()
```



The number of titles released increased significantly from the 2000s onward, with a sharp rise after 2015 and a peak around 2018–2019.

5. SQL Database and Analysis

The prepared dataset was imported into MySQL so that catalogue questions could be answered with SQL. SQL supports filtering, grouping, aggregation, sorting and calculating summary statistics.

Example analysis queries

Count records by content type:

```
SELECT type, count(*) AS total_titles,
round(COUNT(*) * 100 / (SELECT COUNT(*) FROM netflix), 2)
AS percentage FROM netflix GROUP BY type;
```

Count records by release year:

```
SELECT release_year, count(*)  AS total_titles FROM netflix
WHERE release_year IS NOT NULL
GROUP BY release_year
ORDER BY release_year;
```

Average movie duration (when duration values use the “min” suffix):

```
SELECT ROUND(AVG(CAST(REPLACE(duration, ' min', '') AS UNSIGNED)), 2)
AS avg_movie_duration
FROM netflix
WHERE type = 'Movie' AND duration LIKE '%min';
```

MySQL: database/table and successful query

type	total_titles	percentage
Movie	6131	69.62
TV Show	2676	30.38

release_year	total_titles
2006	96
2007	88
2008	136
2009	152
2010	194
2011	185
2012	237
2013	288
2014	352
2015	560
2016	902
2017	1032
2018	1147
2019	1030
2020	953
2021	592

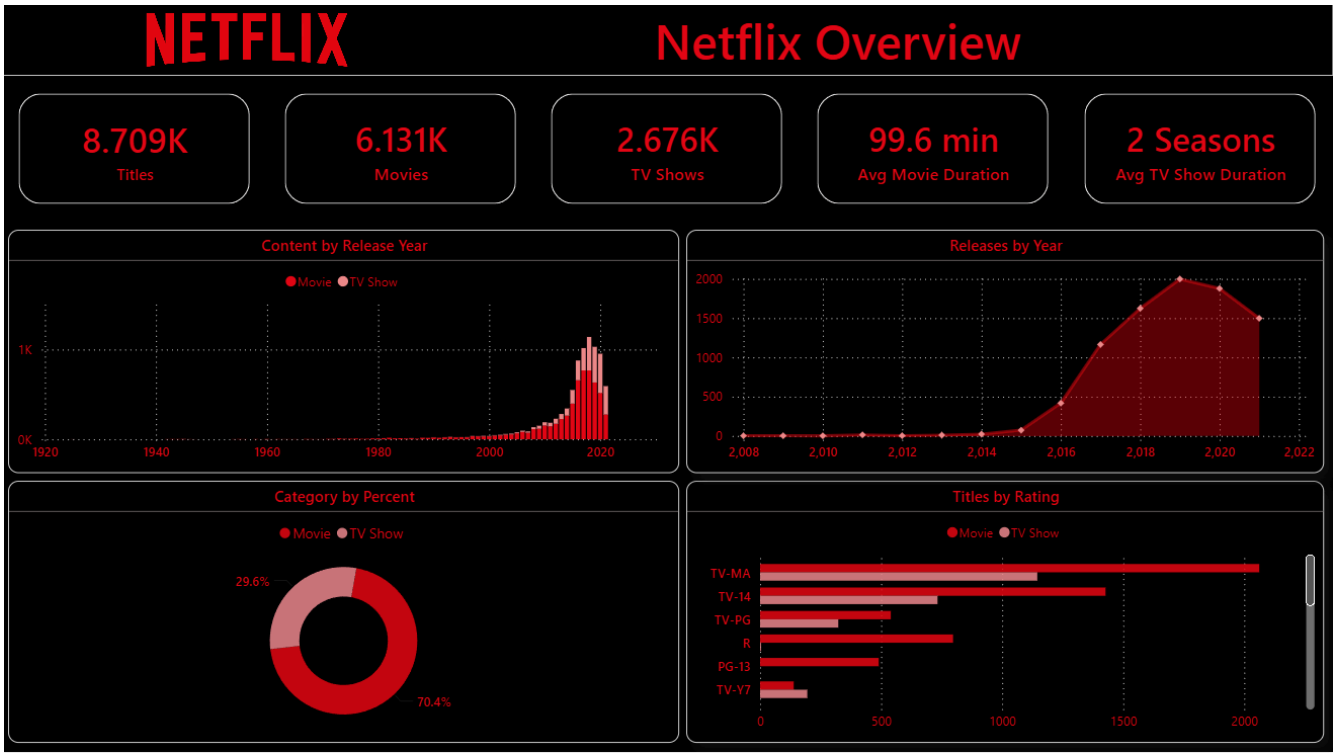
avg_movie_duration
99.58

6. Power BI Dashboard — Page 1

Netflix Overview

The overview page will let a viewer understand the catalogue at a glance. KPI cards include total titles, movie count, TV show count, average movie duration and average TV show duration. Charts include content by release year, distribution by category, titles added by year and titles by rating.

Power BI: Netflix Overview page



• Netflix's catalog contains **8.709K titles**, made up of **6.131K Movies** and **2.676K TV shows**.

• Movies dominate the library at **70.4%**, while TV shows make up **29.6%**, so roughly 7 in 10 titles are movies.

• The average movie runs **99.6 minutes**, while the average TV show runs **2 seasons**, suggesting most series are short-run.

• Content volume is heavily skewed toward recent years: very few titles were released before 2000, followed by a sharp climb through the 2000s and 2010s.

• Releases rise steadily from 2008 and peak around **2018-2020**, with over 1,000 titles in those years, before declining toward 2022.

• **TV-MA** is the most common rating with roughly 1,800+ titles, followed by **TV-14**, **TV-PG**, **R**, and **PG-13**, showing the catalog leans toward mature and teen audiences.

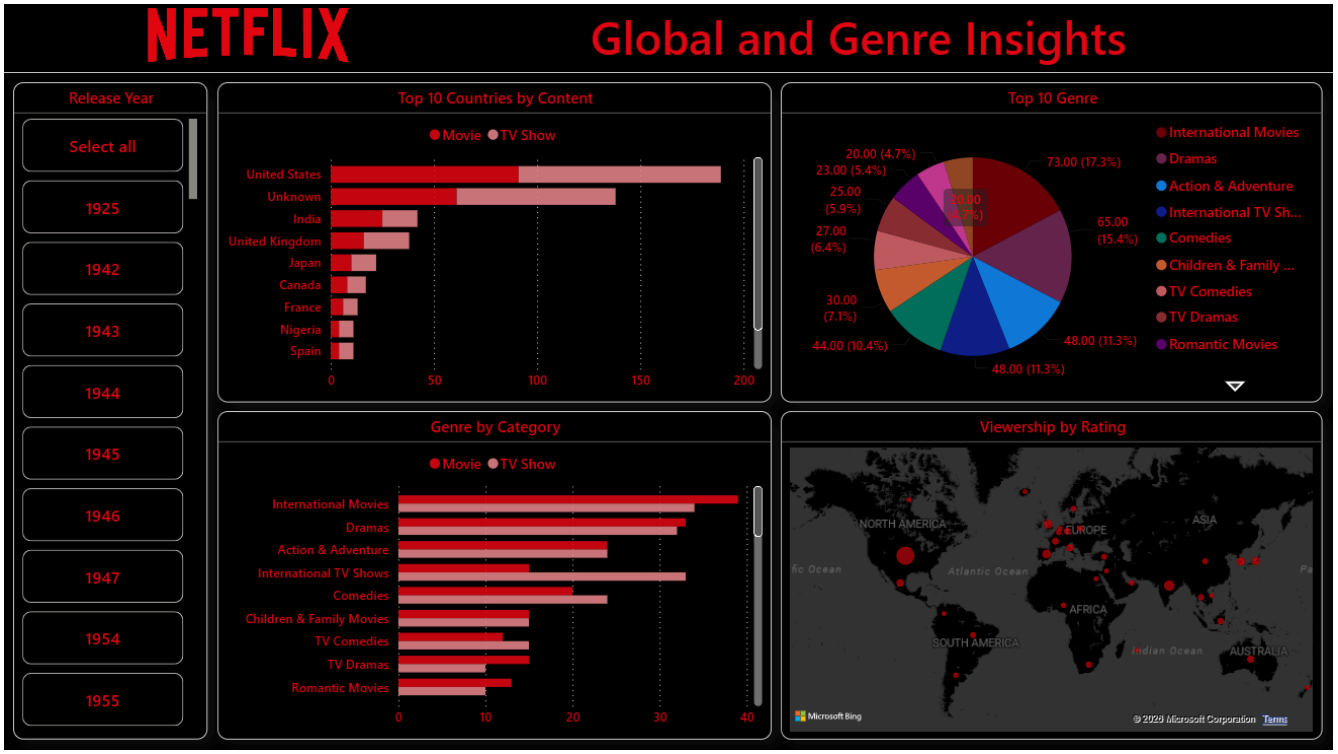
• Movies outnumber TV shows across every rating category, reinforcing the movie-heavy nature of the platform.

7. Power BI Dashboard — Page 2

Global and Genre Insights

This page will explore country and genre metadata, for example top countries, top listed genres, country by content type, genre by content type and viewership by ratings.

Power BI: Global & Genre Insights page



• The **United States** leads all countries in content volume at roughly **190-200 titles**, with a fairly even split between movies and TV shows.

• **"Unknown"** is the second-largest origin at about **140 titles**. This points to missing country data in the dataset, which is worth flagging as a data quality limitation.

• Content production is heavily concentrated: the US alone has more than four times as many titles as India, the next named country.

• In the Top 10 Genre breakdown, the largest genre accounts for **73 titles (17.3%)**, followed by **65 (15.4%)**, then two genres tied at **48 each (11.3%)**.

• The remaining genres range from **44 (10.4%)** down to **20 (4.7%)**, so the top four genres alone make up roughly **55%** of the top-10 total.

• **International Movies** and **Dramas** are the two most common genres, each reaching close to **40 titles** in the Genre by Category chart.

• Movie genres lead the category chart, but **International TV Shows** and **TV Comedies** show that TV content is strongest in international and comedy formats.

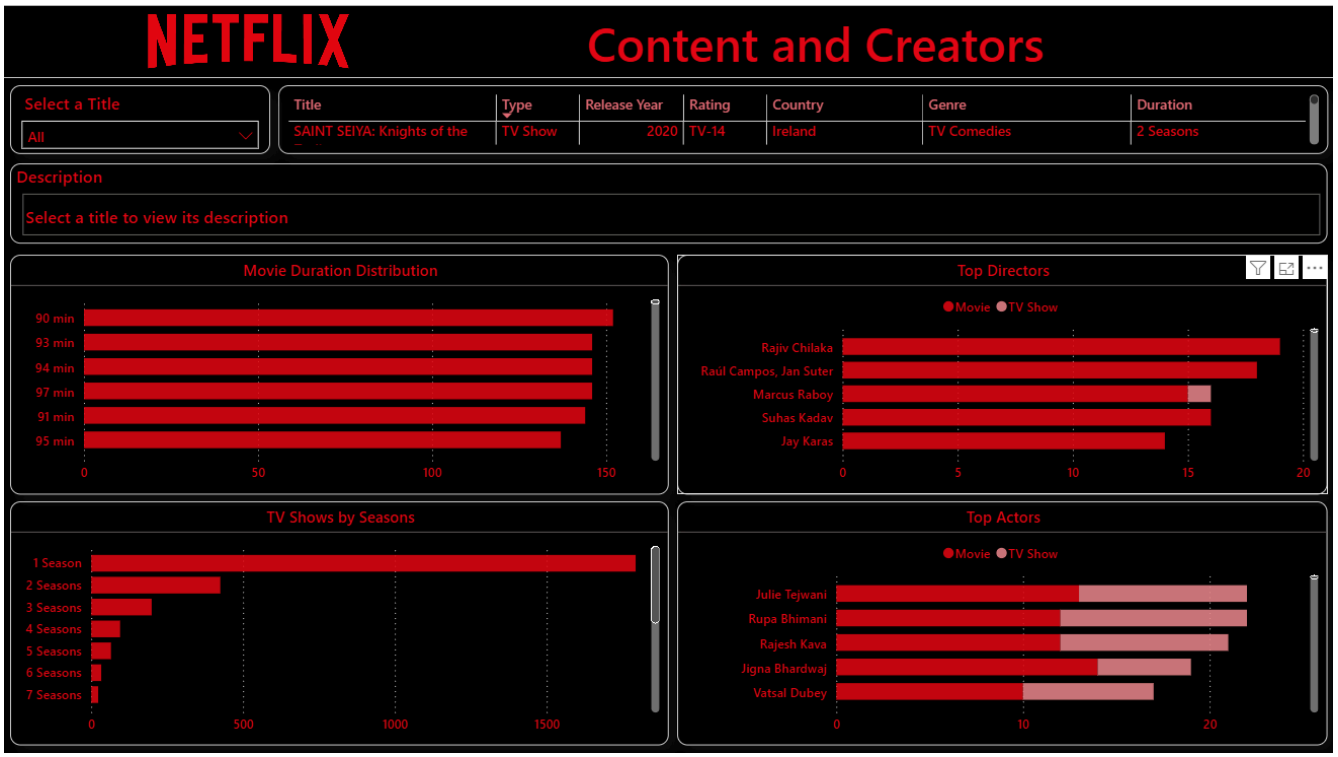
• The world map shows the largest concentration of content in **North America**, with dense clusters across **Europe** and strong presence in **India and Asia.**

8. Power BI Dashboard — Page 3

Content and Creators

This page will show movie-duration distribution, TV-show season counts, and top directors and cast members where the data has been prepared appropriately. A title slicer will let a user choose a title and view its available details, such as type, release year, rating and description.

Power BI: Content & Creators page



• The Content and Creators page lets users select any title from a dropdown to see its type, release year, rating, country, genre, duration, and description in one place.

• The movie duration distribution shows that most films cluster tightly between 90 and 97 minutes, with 90 min the most common runtime at roughly 150 titles. Runtimes of 93, 94, 97, 91, and 95 minutes each follow closely at around 130 to 145 titles.

• TV shows are overwhelmingly short-run: 1 Season dominates at roughly 1,800+ titles, while 2 Seasons falls to around 450 and 3 Seasons to about 250.

• Show counts drop steadily as seasons increase, with 4 to 7 Seasons each representing only a small handful of titles, so long-running series are rare in the catalog.

• Rajiv Chilaka is the top director with about 20 titles, followed closely by the duo Raúl Campos and Jan Suter at roughly 19, then Marcus Raboy, Suhas Kadav, and Jay Karas at around 15 to 17 each.

• The top directors are mostly movie-focused, with only a small TV show share visible for Marcus Raboy.

• The top actors are Julie Tejwani, Rupa Bhimani, Rajesh Kava, Jigna Bhardwaj, and Vatsal Dubey, each appearing in roughly 12 to 20 titles.

• Unlike the directors, the top actors show a balanced mix of movies and TV shows.

9. Key Findings, Limitations and Conclusion

Key findings

1. The dataset contains 8.709K total records: 6.131K Movies and 2.676K TV Shows, so movies make up about 70.4% of the catalogue.

2. The most represented release years in my analysis are 2018-2020, with 1,000+ titles each, after which releases decline toward 2022.

3. The most common rating/category in my analysis is TV-MA, with 1,800+ records, followed by TV-14, which shows the catalogue leans toward mature and teen audiences.

4. One pattern observed in the duration or season-count analysis is that movies run about 90-97 minutes, and most TV Shows are short-run with 1 season being by far the most common.

5. One additional verified observation from the country/genre analysis is that the United States leads all countries in content volume(about 200 titles), and International Movies is the top genre at 73 titles(17.3% of the top 10 genres).

Limitations

• Missing values and inconsistent formatting can affect analysis.

• Fields such as country, cast and listed_in can contain multiple comma-separated values; counts need careful preparation.

• The catalogue dataset does not provide viewership, revenue or user engagement measures.

• Results depend on the dataset version and the cleaning choices made.

Conclusion

This project demonstrates an end-to-end data analytics workflow: inspecting and preparing data with Python, querying structured records with MySQL, and presenting findings through a multi-page Power BI dashboard. It brings together data cleaning, exploratory analysis, SQL aggregation and visual communication in one portfolio project.